Zürich, Switzerland & Oslo, Norway
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Xiangtian Shi

Education

2026 – Present **M.Sc. in Robotics, Systems and Control**, *ETH Zürich*, Switzerland
Current coursework: Model Predictive Control, Robot Dynamics, Planning and Decision Making for Autonomous Robots, Vision Algorithms for Mobile Robotics.

2023 – 2026 **B.Sc. in Vehicle Engineering**, *KTH Royal Institute of Technology*, Sweden (GPA: 4.9/5.0)
Selected coursework: Automatic Control, Optimization, Numerical Methods, Applied Computer Science, Signals and Mechanical Systems, Electrical Engineering.

Technical Skills

Programming Python, MATLAB, Arduino (C++)

Tools PyTorch, Git, Solid Edge, ANSYS, RocketPy

Engineering Control systems, state estimation, embedded systems, mechanical design, hardware integration and troubleshooting, experimental setup design, data acquisition

Technical Experience and Projects

06.2026 – Present **Biomimetic Flapping-Wing Robot Sim-to-Real Research**
Visiting Student Researcher, Dr. Tao Du's Group
Institute for Interdisciplinary Information Sciences (IIIS), Tsinghua University, China
- Developed an ESP32-based wireless SBUS control interface for the group's flapping-wing research; integrating and troubleshooting experimental hardware.
- Redesigned mechanical components for motion-capture and force-measurement data acquisition to improve measurement reliability and support sim-to-real validation.
- Contributed to simulation workflows connecting model development with physical experiments.

01.2026 – 06.2026 **PINNs for Optimal Trajectory Planning of Unicycle Robots**
Bachelor's Thesis, KTH Royal Institute of Technology, Sweden
- Developed PyTorch frameworks for physics-informed trajectory optimization under kinematic and control constraints.
- Implemented physics-based constraint enforcement and obstacle-avoidance strategies for unicycle trajectory planning.
- Evaluated loss balancing and constraint enforcement across the optimization workflow.

01.2025 – 09.2025 **Rocket Avionics Engineer**
AESIR Student Rocketry Association, KTH, Sweden
- Refined the simulation model and reduced apogee-prediction error from 10% to 1.3% in RocketPy reference simulations.
- Designed and tuned a PID controller for active stabilization in RocketPy, achieving simulated altitude control within 2.7% of the 3,000 m target.

01.2024 – 06.2024 **Solar-Powered Model Vehicle**, *KTH Project Course*, Sweden
- Designed and built a vehicle integrating solar power, mechanical components, and electrical subsystems.
- Implemented Arduino-based control and evaluated the integrated system through performance testing.

Work Experience

10.2025 – 06.2026 **Teaching Assistant in Electrical Engineering**, *Unit of Mechatronics, KTH*, Sweden

- Supported laboratory instruction in circuit theory, signal processing, and mechatronics.
- Evaluated assignments and assisted with course coordination.

05.2023 – 01.2025 **Mathematics Tutor**, *MentorNorge AS*, Norway

Delivered individualized mathematics tutoring adapted to each student's learning needs.

10.2022 – 06.2023 **Kindergarten Assistant**, *Høvik kindergarten*, Norway

Supported children aged 2–4 in daily routines and structured creative and educational activities.

Extracurricular Activities

09.2023 – 06.2024 **International Group**, *KTH Student Council*, Sweden

Organized social and cultural events supporting integration among international students.

10.2020 – 08.2022 **Competitive Esports**, *PUBG Mobile team competition*

Won the PUBG Mobile Pro League Western Europe Spring 2021 in a 16-team professional field and competed internationally.

Languages

Norwegian (Native); Chinese (Native); English (Fluent; IELTS Academic 7.5); Swedish (Fluent; CEFR B2)

Hobbies and Interests

Violin (10+ years, regional competition winner); Table tennis (club member, regional competitions); Running (Stockholm Half Marathon 2025); Hiking; Skiing